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Practical 09

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

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Semester / Year:4th/2nd

Academic Session:

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CODE :-

```
M ~  
Shweta Bambode@Shweta MSYS ~  
$ nano opt.c
```

```
M -  
GNU nano 8.7 opt.c  
#include <stdio.h>  
  
int main() {  
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};  
    int frames[3];  
    int n = 12, frames_count = 3;  
    int i, j, k, pos, farthest;  
    int hit = 0, fault = 0;  
    int flag;  
  
    for(i = 0; i < frames_count; i++)  
        frames[i] = -1;  
  
    for(i = 0; i < n; i++) {  
        flag = 0;  
  
        // Check for page hit  
        for(j = 0; j < frames_count; j++) {  
            if(frames[j] == pages[i]) {  
                hit++;  
                flag = 1;  
                break;  
            }  
        }  
    }  
}
```

```
GNU nano 8.7 opt.c

// Page fault
if(flag == 0) {
    pos = -1;
    farthest = i + 1;

    for(j = 0; j < frames_count; j++) {
        int found = 0;
        for(k = i + 1; k < n; k++) {
            if(frames[j] == pages[k]) {
                if(k > farthest) {
                    farthest = k;
                    pos = j;
                }
                found = 1;
                break;
            }
        }

        if(found == 0) {
            pos = j;
            break;
        }
    }

    if(pos == -1)
        pos = 0;

    frames[pos] = pages[i];
    fault++;
}
}
```

```
float hit_percent = (hit * 100.0) / n;
float fault_percent = (fault * 100.0) / n;

printf("Total Page Hits: %d\n", hit);
printf("Total Page Faults: %d\n", fault);
printf("Page Hit Percentage: %.2f%%\n", hit_percent);
printf("Page Fault Percentage: %.2f%%\n", fault_percent);

return 0;
}
```

OUTPUT:-

```
Shweta Bambode@Shweta MSYS ~  
$ gcc opt.c -o opt  
  
Shweta Bambode@Shweta MSYS ~  
$ ./opt  
Total Page Hits: 5  
Total Page Faults: 7  
Page Hit Percentage: 41.67%  
Page Fault Percentage: 58.33%
```